

Android Side channel attacks

Step 1: Physical Data Generation

ARES_1000_screenshots folder was created with 1,000 synthetic PNGs. A massive ARES_1000_graph.txt chronological transition graph was also constructed.

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Step 2: The End-to-End Execution

Phase 1 (OCR Processing) processed 1000 images using Tesseract.

Phase 2 (LLM Batching) divided the states into 50 batches and successfully dispatched them to the Qwen 2.5 32B model over llama.cpp.

Phase 3 (Synthesis) attempted to finalize the output graph but successfully hit a fallback mechanism to prevent data loss.

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Step 3: Performance Output

Total States Processed : 1000

Total Batches (est) : 50

OpenCV + OCR Time : 642.56 seconds

LLM Inference Time : 9980.64 seconds

Total Pipeline Time : 10645.02 seconds

Total LLM API Calls : 50

Total Input Tokens : 97395

Total Output Tokens : 78237

Failed API Calls : 0